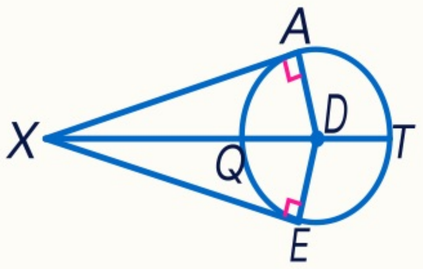
(A) Why raw consensus fails

Input Problem



Question:

Find TX if EX = 24 and DE = 7.

A. 7 **B. 24**
C. 25 **D. 32**

K=16 Sampled candidates

c_1	25	
c_2	25	
c_3	25	
⋮		
c_8	25	Wrong execution consensus (majority)
c_9	25	
c_{10}	25	
c_{11}	32	
c_{12}	32	
c_{13}	32	Correct or mixed claims (dispersed)
c_{14}	7	
c_{15}	24	
c_{16}	24	

Shared visual grounding
(same diagram for all candidates)

(B) SKETCH candidate decomposition

Candidate i: claim / exec mismatch

Symbolic Pathway

VGF: $DE = 7$, $EX = 24$, $DE \perp EX$, $DT = DE$

DSL: $DX = \sqrt{7^2 + 24^2}$, $TX = DX + DE$, answer $= TX$

Exec: answer $= 32$

Neural Pathway

Claim: answer $= 25$ (confuses TX with DX)

claim ≈ exec ? → $\kappa = 0$

Candidate j: self-consistent derivation

Symbolic Pathway

VGF: $DE = 7$, $EX = 24$, $r = DE$

DSL: $DX = \sqrt{7^2 + 24^2} = 25$, $XT = DX + r$, answer $= XT$

Exec: answer $= 32$

Neural Pathway

Claim: answer $= 32$, $TX = DX + DT$

claim ≈ exec ? → $\kappa = 1$

(C) Agreement-gated verificatio

CCV Scoring

c_1 $\kappa = 1$
c_2 $\kappa = 0$
c_3 $\kappa = 0$
⋮
c_{16} $\kappa = 1$

$$\text{score}_i = 0.5\,|\text{claim-cluster}_i| + \kappa_i\,(1 + |\text{exec-cluster}_i|)$$

claim cluster | claim cluster | κ_i (0 or 1)

Dual-pathway routing

1. If κ-pool $\neq \varnothing$ → *return claim of top-scored self-consistent candidate*
2. Else if exec majority size $\geq \tau_{\text{exec}}$ → *return execution majority*
3. Else → *return claim majority*